

Mathematics — Exam 04 (Ch 4)

SINGLE ATTEMPT

Factorization & Algebraic Manipulation | Student: Seemab | Date: 25 May 2026 | Marks: 65 | Version: 2 (Corrected)

Corrected 2026-05-25: MCQ re-read from higher-quality screenshots (was 5/12, corrected to 12/12). Total raised from 51/65 (78.5%) to 57/65 (87.7%). Grade A.

TOTAL	MCQ	SECTION B	SECTION C	GRADE
57/65	12/12	26/33	19/20	A
87.7%	100%	78.8%	95.0%	87.7%

Section Breakdown

Section A (MCQ)	100%	12/12
Section B (Short)	78.8%	26/33
Section C (Long)	95.0%	19/20
Total	87.7%	57/65

Performance Card

Strengths this attempt

- MCQ perfect 12/12 — all definitional and identity questions correct
- Substitution factorization: Q.3(i) 5/5, flawless
- HCF of three polynomials: Q.3(ii) 5/5, lowest-power rule applied precisely
- Algebraic fractions with sign-change: Q.3(iv) 5/5
- Square root by factorization: Q.2(xi) 3/3
- Division of algebraic fractions: Q.2(x)(or) 3/3
- HCF by factorization (both Q.2(vii) and Q.2(viii)(or)): full marks

Areas for revision

- **Sum-of-cubes middle term:** $-ab$ not $-2ab$; Q.2(v) wrote $-6xy$ instead of $-3xy$
- **Sophie Germain identity step 2:** Q.2(iii) stopped at $(x^2+2m^2)^2 - (2xm)^2$ without applying difference-of-squares
- **Grouping to difference-of-squares:** Q.2(ii)(or) had $1 - (x+y)^2$ but did not factor further
- **Sign errors in expansion:** Q.2(vi)(or) wrote $-150x$ instead of $+150x$, preventing recognition of $(2x-5)^3$
- **LCM of three trinomials:** Q.2(ix)(or) — none of the three trinomials correctly factored
- **Verification step layout:** Q.3(iii) — messy intermediate steps cost 1 mark despite the correct conclusion

Section B breakdown of the 7-mark drop (33 – 26 = 7):

- Q.2(ii)(or): -1 (final factored form not reached from recognized setup)
- Q.2(iii): -1 (Sophie Germain stopped one step early)
- Q.2(v): -1 ($-6xy$ instead of $-3xy$ in trinomial factor)
- Q.2(vi)(or): -2 (sign error on $150x$; $(2x-5)^3$ never reached)
- Q.2(ix)(or): -2 (trinomials not factored; LCM line meaningless)

Every deduction has a specific, learnable fix. None of these is a concept Seemab does not understand; they are all execution errors under time pressure.

Comparison: All Math Attempts

Exam	Date	MCQ	Sec B	Sec C	Total	%	Change
exam-01-ch1-2	18 Apr 2026	12/12	33/33	20/20	65/65	100%	—
exam-03-ch4-5-7	22 Apr 2026	12/12	28.5/33	20/20	60.5/65	93.1%	−6.9%
exam-04-ch4 (this)	25 May 2026	12/12	26/33	19/20	57/65	87.7%	−5.4%

MCQ streak intact across all three Math exams (and all 12 exams to date). Section B is the variable element: 33, 28.5, 26 — formula application consistency is the lever to pull. Section C held up well at 19/20. The 87.7% is a solid Grade A; the path to 93%+ on the next Ch4 exam is specifically through the five Section B error types listed above.